

OLA Ride Insights

Domain:

Ride-Sharing & Mobility Analytics

1. Objective:

The objective of this project is to analyze OLA ride booking data and generate meaningful business insights using SQL, SQLite, Python, and Streamlit. The project focuses on understanding ride trends, customer behaviour, revenue patterns, cancellation reasons, and service quality through ratings analysis.

The final outcome is an interactive Streamlit dashboard that enables users to explore ride-sharing data and make data-driven decisions.

2. Problem Statement

Ride-sharing platforms generate large volumes of data related to bookings, payments, customer ratings, driver performance, and cancellations.

Analyzing this data can help businesses:

- Improve operational efficiency
- Increase customer satisfaction
- Optimize revenue generation
- Reduce ride cancellations
- Understand customer preferences

This project aims to transform raw ride data into actionable business insights through data analysis and visualization.

3. Dataset Description

The dataset contains ride booking information from OLA.

Key Attributes

- Booking ID
- Customer ID

- Vehicle Type
 - Booking Status
 - Booking Value
 - Ride Distance
 - Payment Method
 - Customer Rating
 - Driver Rating
 - Cancellation Reasons
 - Incomplete Ride Reasons
-

4. Data Cleaning and Preprocessing

The dataset underwent preprocessing to improve data quality and consistency.

Steps Performed

- Handled missing values using appropriate techniques.
- Standardized column names.
- Dropped two columns V_TAT, C_TAT.
- Validated booking status values.
- Verified payment method information.
- Checked ride distance and booking value consistency.
- Removed data inconsistencies.
- Performed exploratory data analysis (EDA).
- Prepared cleaned data for SQLite storage and analysis.

Outcome

A clean and structured dataset was created for SQL querying and dashboard development.

5. Database Design

Database Used

olaride_db.sqlite

Table Name

ride_data

SQLite was chosen because it is lightweight, portable, and suitable for dashboard deployment through Streamlit Cloud.

6. SQL Analysis

The following SQL queries were developed to extract business insights:

Query 1

Retrieve all successful bookings.

```
"1. Retrieve all successful bookings":  
""
```

```
SELECT Booking_ID,Booking_Status  
FROM ride_data  
WHERE Booking_Status = 'Success'  
""
```

Choose a Query

1. Retrieve all successful bookings

Query Result

	Booking_ID	Booking_Status
0	CNR2940424040	Success
1	CNR2982357879	Success
2	CNR1797421769	Success
3	CNR8787177882	Success
4	CNR3612067560	Success
5	CNR4787583516	Success
6	CNR7943634301	Success
7	CNR4524472111	Success
8	CNR8181602032	Success
9	CNR8090918544	Success

Business Insight: This analysis helps understand total completed ride volume and overall platform performance efficiency.

Query 2

Find average ride distance for each vehicle type.

"2. Average ride distance for each vehicle type":
""

```
SELECT
    Vehicle_Type,
    ROUND(AVG(Ride_Distance), 2) AS Avg_Ride_Distance
FROM ride_data
GROUP BY Vehicle_Type
ORDER BY Avg_Ride_Distance DESC
""
```

Choose a Query

2. Average ride distance for each vehicle type

Query Result

	Vehicle_Type	Avg_Ride_Distance
0	Prime Sedan	15.76
1	eBike	15.58
2	Bike	15.53
3	Mini	15.51
4	Prime Plus	15.45
5	Prime SUV	15.27
6	Auto	6.24

Business Insight: This helps identify usage patterns across vehicle types and supports fleet optimization and pricing strategy decisions.

Query 3

Calculate total customer cancellations.

"3. Total number of rides cancelled by customers":
""

```
SELECT
    Booking_Status,
    COUNT(*) AS Total_Cancelled_Rides
FROM ride_data
WHERE Booking_Status = 'Canceled by Customer'
GROUP BY Booking_Status
""
```

 **OLA RIDE INSIGHTS SQL Query Results**

Choose a Query

3. Total number of rides cancelled by customers

Query Result

	Booking_Status	Total_Cancelled_Rides
0	Canceled by Customer	10499

Business Insight: This highlights customer cancellation behavior and helps identify issues such as pricing, waiting time, or service experience.

Query 4

Identify top 5 customers by ride count.

"4. Top 5 customers by number of rides":

"""

```
WITH customer_rides AS (  
    SELECT  
        Customer_ID,  
        COUNT(*) AS Total_Rides,  
        ROW_NUMBER() OVER (  
            ORDER BY COUNT(*) DESC  
        ) AS rn  
    FROM ride_data  
    GROUP BY Customer_ID  
)
```

```
SELECT *  
FROM customer_rides  
WHERE rn <= 5
```

"""

Choose a Query

4. Top 5 customers by number of rides

Query Result

	Customer_ID	Total_Rides	rn
0	CID954071	5	1
1	CID980727	4	2
2	CID969725	4	3
3	CID966929	4	4
4	CID952434	4	5

Business Insight: This identifies high-value customers who can be targeted for loyalty programs and personalized offers.

Query 5

Determine rides cancelled due to driver-related issues.

"5. Driver cancellations due to personal and car-related issues":

"""

```
SELECT  
    COUNT(*) AS Driver_Cancelled_Rides  
FROM ride_data  
WHERE Canceled_Rides_by_Driver =  
'Personal & Car related issue'
```

"""

Choose a Query

5. Driver cancellations due to personal and car-related issues

Query Result

	Driver_Cancelled_Rides
0	6542

Business Insight: This helps detect operational issues related to driver availability and vehicle condition, improving service reliability.

Query 6

Find maximum and minimum driver ratings for Prime Sedan bookings.

```
"6. Maximum and minimum driver ratings for Prime Sedan":
"""
```

```
SELECT
|   MAX(Driver_Ratings) AS Max_Driver_Rating,
|   MIN(Driver_Ratings) AS Min_Driver_Rating
FROM ride_data
WHERE Vehicle_Type = 'Prime Sedan'
"""
```

Choose a Query

6. Maximum and minimum driver ratings for Prime Sedan

Query Result

	Max_Driver_Rating	Min_Driver_Rating
0	5	3

Business Insight: This helps evaluate driver performance quality and identify service consistency within the Prime Sedan category.

Query 7

Calculate total rides paid using UPI.

```
"7. Total rides paid using UPI":
"""
```

```
SELECT
|   COUNT(*) AS Total_UPI_Rides
FROM ride_data
WHERE Payment_Method = 'UPI'
"""
```

Choose a Query

7. Total rides paid using UPI

Query Result

	Total_UPI_Rides
0	25881

Business Insight: This shows digital payment adoption trends and customer preference for cashless transactions.

Query 8

Find average customer rating by vehicle type.

```
"8. Average customer rating per vehicle type":  
""
```

```
SELECT  
    Vehicle_Type,  
    ROUND(  
        AVG(Customer_Rating),  
        2  
    ) AS Avg_Customer_Rating  
FROM ride_data  
GROUP BY Vehicle_Type  
ORDER BY Avg_Customer_Rating DESC  
""
```

Choose a Query

8. Average customer rating per vehicle type

Query Result

	Vehicle_Type	Avg_Customer_Rating
0	Prime Plus	4.01
1	Prime Sedan	4
2	Prime SUV	4
3	Mini	4
4	Bike	4
5	Auto	4
6	eBike	3.99

Business Insight: This helps compare customer satisfaction across vehicle types and identify best-performing categories.

Query 9

Calculate total booking value from successful rides.

```
"9. Total booking value of successful rides":  
""
```

```
SELECT  
    SUM(Booking_Value)  
    AS Total_Booking_Value  
FROM ride_data  
WHERE Booking_Status = 'Success'  
""
```

Choose a Query

9. Total booking value of successful rides

Query Result

	Total_Booking_Value
0	35080467

Business Insight: This provides insights into total revenue generated from completed rides and overall business performance.

Query 10

Identify incomplete rides and their reasons.

"10. Incomplete rides and their reasons":
""

```
SELECT
    Incomplete_Rides_Reason,
    COUNT(*) AS Total_Rides
FROM ride_data
WHERE Incomplete_Rides = 'Yes'
GROUP BY Incomplete_Rides_Reason
ORDER BY Total_Rides DESC
""
```

Choose a Query

10. Incomplete rides and their reasons

Query Result

	Incomplete_Rides_Reason	Total_Rides
0	Customer Demand	1601
1	Vehicle Breakdown	1591
2	Other Issue	734

Business Insight: This helps identify failure points in the booking journey and supports improving ride completion rates and service quality.

7. Streamlit Dashboard Development

An interactive Streamlit application was developed to present insights visually.

Dashboard Pages

Project Introduction

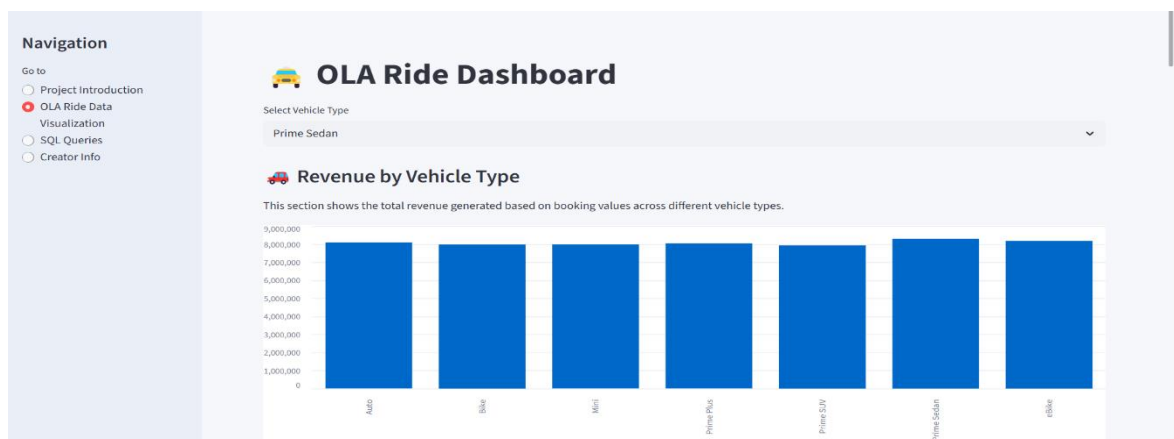
Provides project overview.



OLA Ride Dashboard

Includes:

- Revenue Analysis



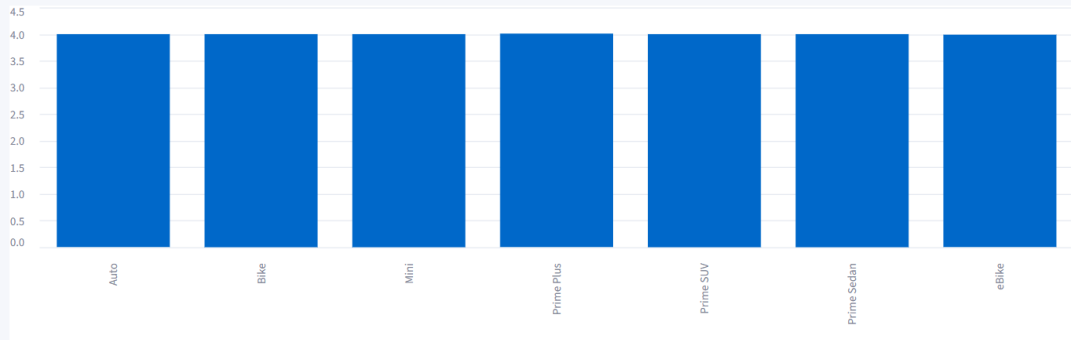
- Booking Status Analysis

Booking Status Distribution		
Shows the count of rides by booking status such as Success, Cancelled, Driver Not Found, and Incomplete.		
	Booking_Status	Total
0	Canceled by Customer	10499
1	Canceled by Driver	18434
2	Driver Not Found	10124
3	Success	63967

- Customer Rating Analysis

★ Customer Rating Analysis

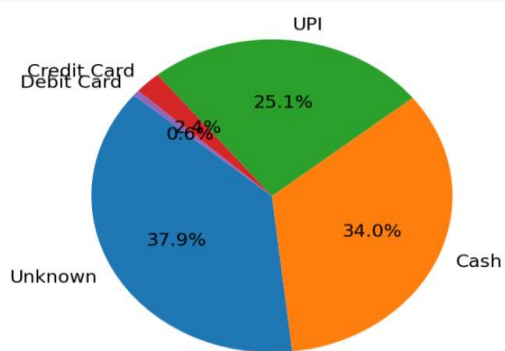
This analysis provides insights into customer satisfaction by studying rating patterns across rides, helping identify service quality trends and improvement areas.



- Payment Method Distribution

💳 Payment Method Distribution

This analysis shows the distribution of payment methods used by customers across rides, helping to understand customer preferences such as cash, digital wallets, or card payments. It provides insights into transaction behavior and supports decisions for improving payment options and digital adoption strategies.



- Outlier Analysis

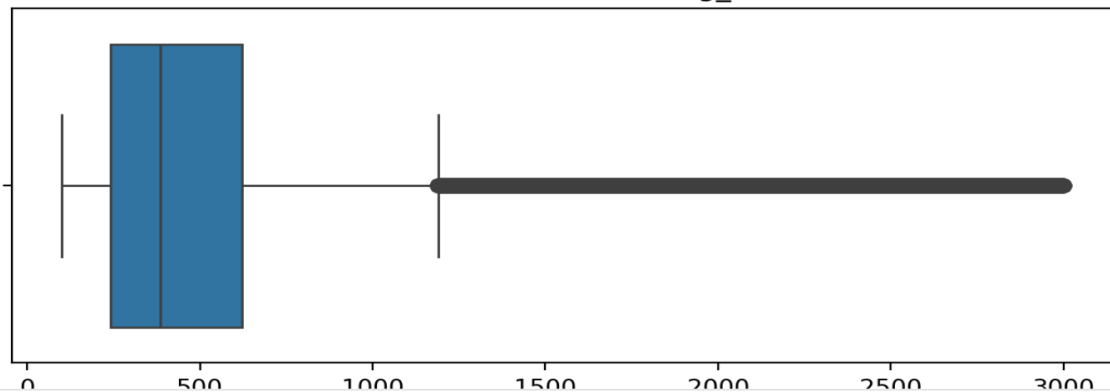
📦 Outlier Analysis

This analysis identifies unusual or extreme values in ride data such as fare amount, distance, or ride duration. Detecting outliers helps highlight potential data errors, fraud cases, or rare high-value trips, and supports improving data quality and operational decision-making.

Select Numeric Column

Booking_Value

Outlier Detection - Booking_Value



SQL Query Results

Displays outputs from SQL queries and associated business insights.

Navigation

Go to

- ☐ Project Introduction
- ☐ OLA Ride Data Visualization
- ☒ SQL Queries
- ☐ Creator Info



OLA RIDE INSIGHTS SQL Query Results

Choose a Query

5. Driver cancellations due to personal and car-related issues

Query Result

Driver_Cancelled_Rides
0

Business Insight: This helps detect operational issues related to driver availability and vehicle condition, improving service reliability.

Creator Information

Project author details and skills.

Navigation

Go to

- ☐ Project Introduction
- ☐ OLA Ride Data Visualization
- ☐ SQL Queries
- ☒ Creator Info



Creator of this Project

Developed by: A Harini

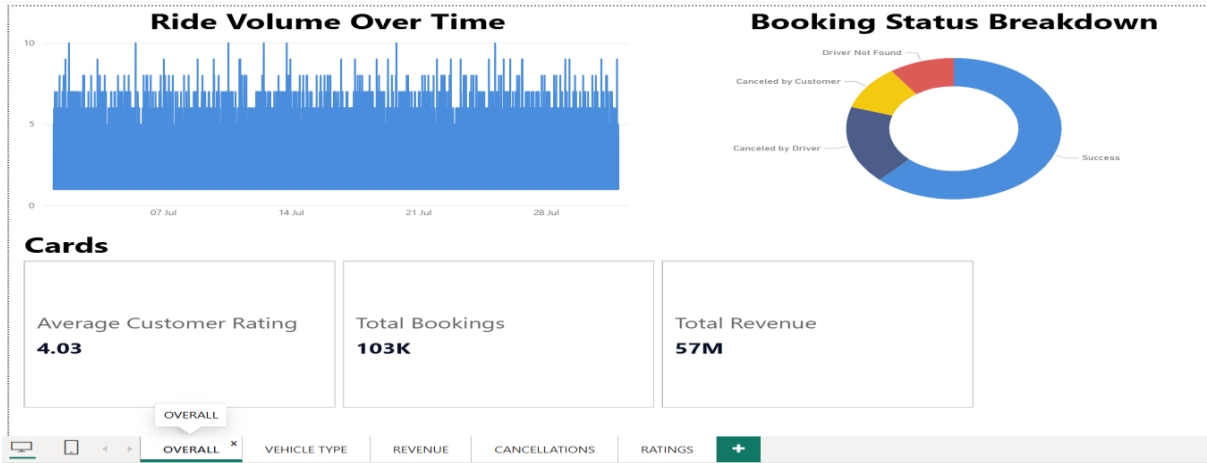
Skills: Python, SQL, Data Analysis,Data Visualization, Pandas, EDA, Matplotlib, Streamlit

8.Power BI

1. Overall

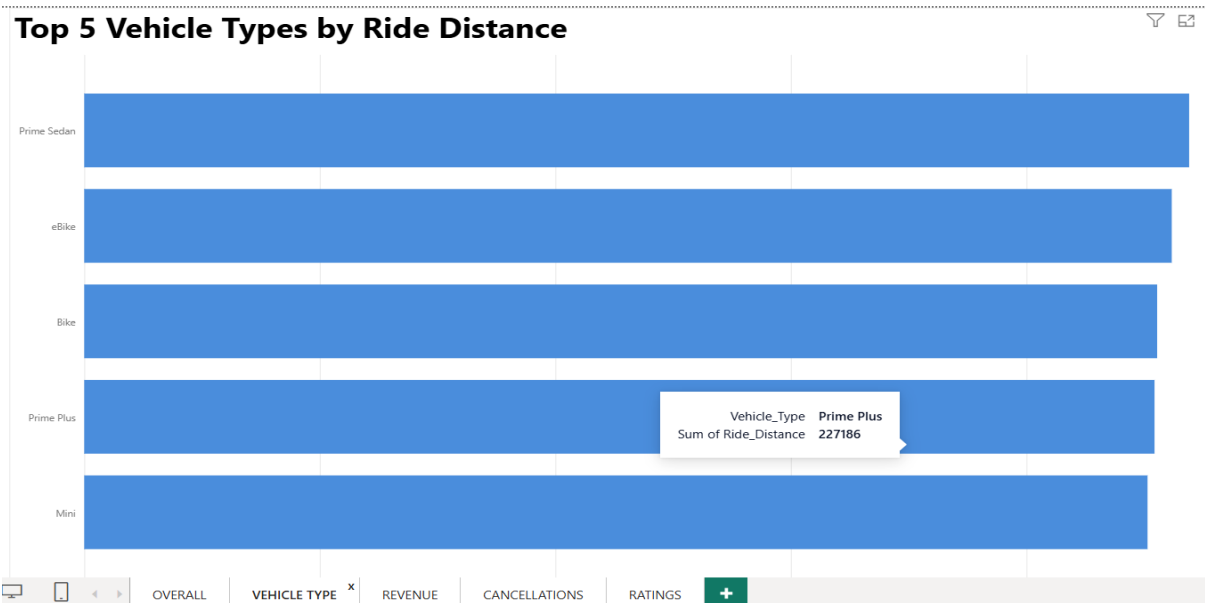
- Ride Volume Over Time

- Booking Status Breakdown



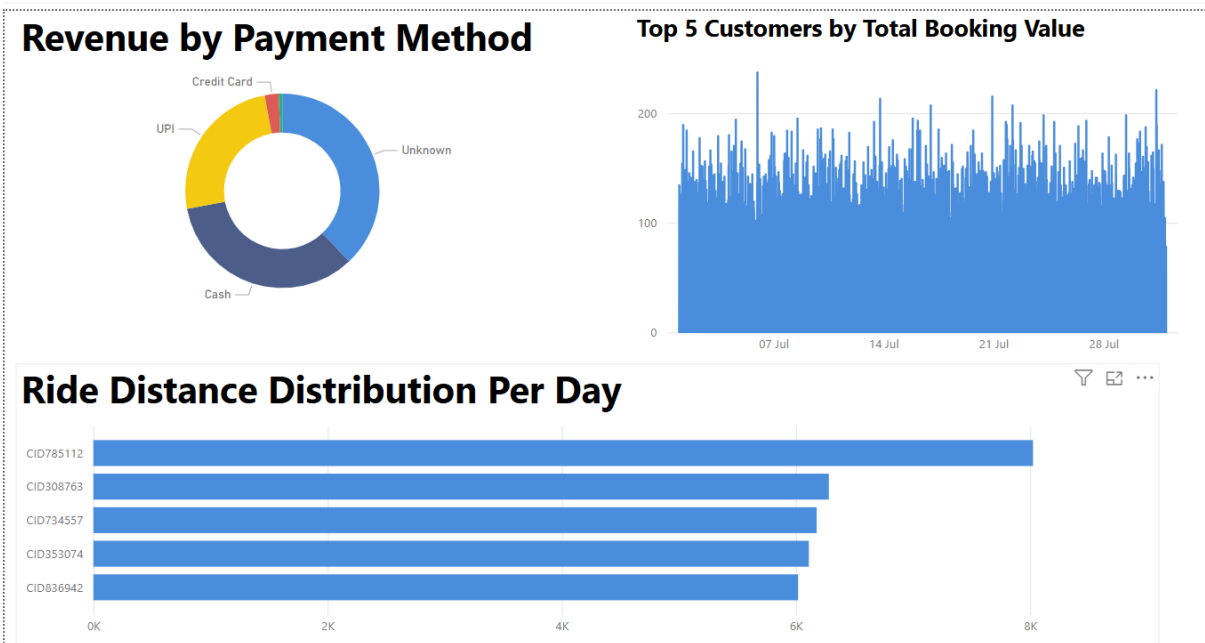
2. Vehicle Type

- Top 5 Vehicle Types by Ride Distance



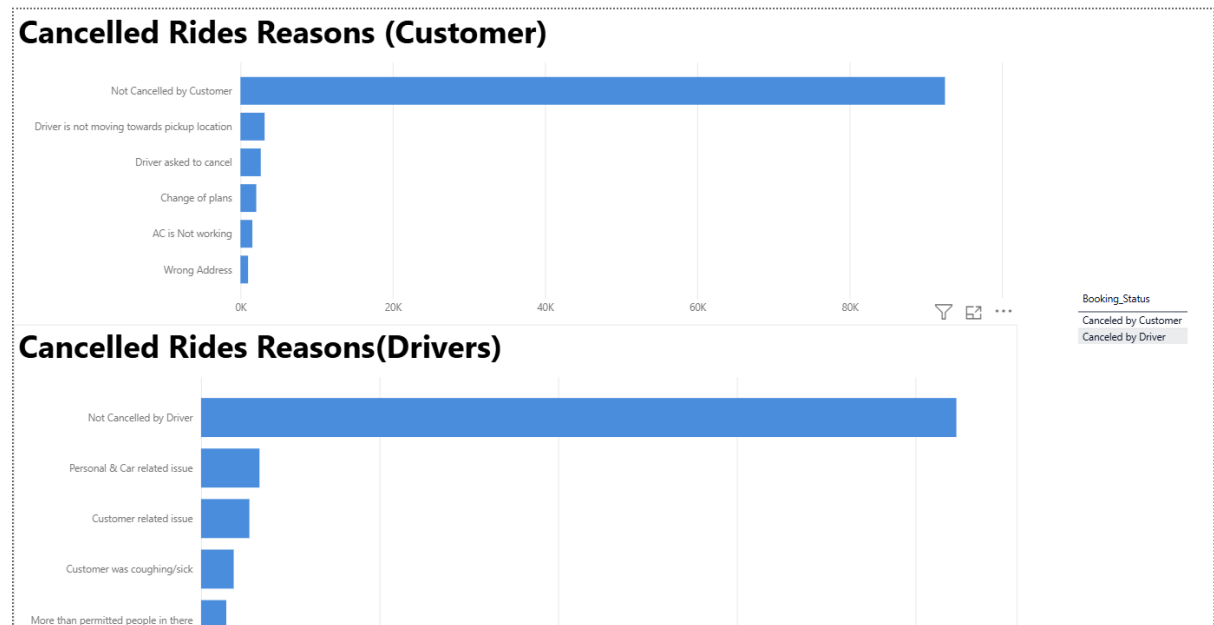
3. Revenue

- Revenue by Payment Method
- Top 5 Customers by Total Booking Value
- Ride Distance Distribution Per Day



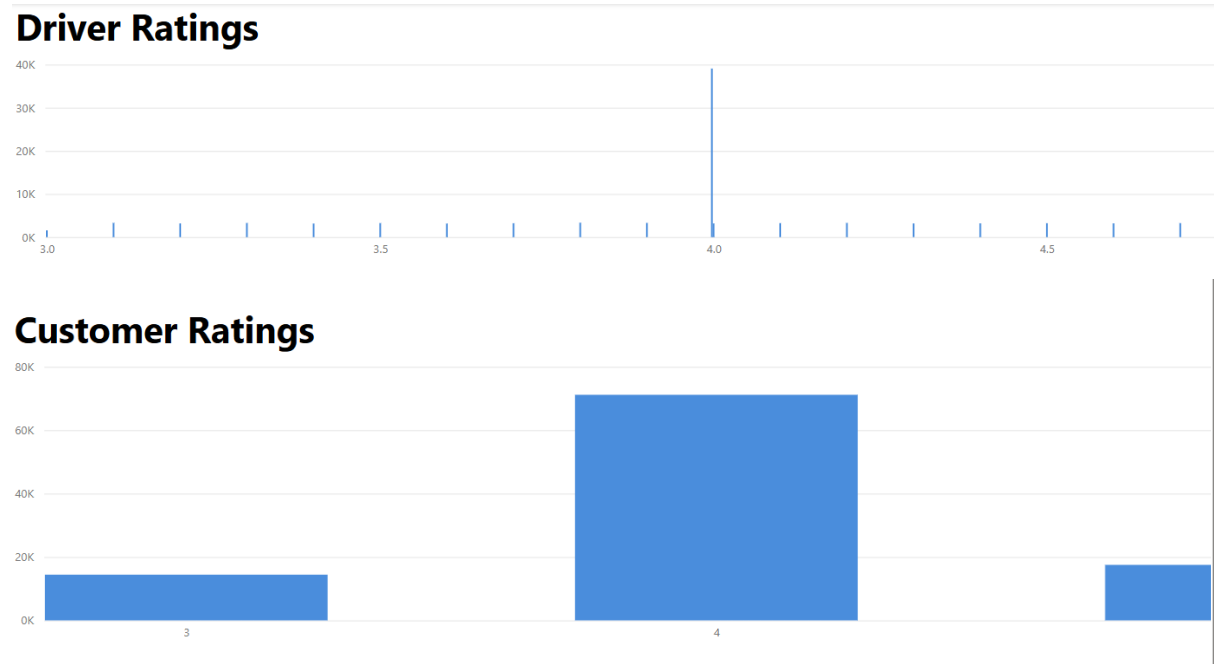
4. Cancellation

- Cancelled Rides Reasons (Customer)
- Cancelled Rides Reasons(Drivers)



5. Ratings

- Driver Ratings
- Customer Ratings



9. Business Insights

The analysis generated the following insights:

- Successful rides account for the majority of bookings.
 - Certain vehicle types generate significantly higher revenue.
 - Digital payment methods are widely preferred.
 - Customer satisfaction varies across vehicle categories.
 - Cancellation patterns reveal operational challenges.
 - High-value customers can be targeted through loyalty programs.
 - Driver ratings help identify service quality trends.
-

10. Results

The project successfully delivered:

- Cleaned and structured ride-sharing dataset.
 - SQLite database for efficient querying.
 - SQL-based business insights.
 - Interactive Streamlit dashboard.
 - Visual representation of ride trends and revenue patterns.
 - Actionable recommendations for operational improvements.
-

11. Future Enhancements

Potential future improvements include:

- Ride demand forecasting using Machine Learning.
- Customer churn prediction.
- Real-time ride monitoring.
- Geographic ride heatmaps.
- Driver allocation optimization.

- Cloud database integration.

12. Technologies Used

Technology	Purpose
Python	Data Analysis
SQL	Data Querying
SQLite	Database Management
Streamlit	Dashboard Development
Pandas	Data Manipulation and Analysis
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
GitHub	Version Control and Project Hosting

14. Project Structure

```
OLA_RIDE_PROJECT/  
|  
├─ OLA_APP.py  
├─ README.md  
├─ OLA_LOGO.png  
├─ OLA_RIDE_ANALYSIS.ipynb  
├─ Python_Sql_Connector.ipynb  
├─ olaride_db.sqlite  
├─ OLA_DataSet.xlsx  
├─ ola_ride_cleaned_data.xlsx  
|  
├─ Report/  
|   └─ OLA_Ride_Insights_Report.pdf
```

```
|  
└─ Screenshots/  
    └─ Dashboard_Home.png  
    └─ Revenue_By_Vehicle_Type.png  
    └─ Booking_Status_Distribution.png  
    └─ Customer_Rating_Analysis.png  
    └─ Payment_Method_Distribution.png  
    └─ Outlier_Analysis.png  
    └─ Outlier_Summary.png  
    └─ SQL_Query_Results.png
```

13. Conclusion

The OLA Ride Insights project demonstrates how data analytics can be used to transform raw ride-booking data into meaningful business intelligence. Through SQL analysis, visualization, and dashboard development, the project provides valuable insights into ride trends, revenue generation, customer behaviour, and service performance, supporting informed decision-making in ride-sharing operations.

Author

A. Harini

Skills: Python, SQL, SQLite, Streamlit, Data Analysis, Data Visualization, EDA, Pandas.

This format is suitable for GitHub, project submission, and portfolio presentation.